

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

University Theory Examination May 2017

PH326.6 STERILE PRODUCT TECHNOLOGY-I

Date: 10.05.2017, Wednesday Time: 10:00 a.m. To 12:00 p.m. (2 hrs) Maximum Marks: 50

Instructions:

- 1. Figures to right indicate marks.**
 - 2. Section wise separate answer book to be used.**
 - 3. Draw neat sketches wherever necessary.**
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SECTION- I

Q 1	.Answer the following:	05
A	What do you mean by terminal sterilization?	01
B	Name all the finished product quality control test for parenterals.	01
C	Write full form of HEPA,HVAC system.	01
D	Define 'Clean room' for parenteral manufacturing facility.	01
E	What is aseptic filling in sterile production?	01
Q 2	Attempt any <u>TWO</u> of the following	
A	Draw a neat labelled diagram of large volume parenteral.	05
B	Note on Laminar air flow hoods.	05
C	Give air classification as per Schedule M for parenteral facility.	05
Q 3	Attempt any <u>TWO</u> of the following	
A	Describe general requirements for the design of aseptic areas.	05
B	Explain men and material entry in sterile production.	05
C	Enumerate different methods of sterilization and explain about Steam Sterilization in detail.	05

SECTION- II

- Q 4 Answer the following:** **05**
- A Give difference between Intradermal and Subcutaneous route of administration. **01**
- B What is pass box in sterile manufacturing? **01**
- C HEPA filter integrity testing (smoke testing) done _____ **01**
1. Yearly 2. 6 Monthly 3. Daily 4. 3 Monthly
- D Name a bacteria which should be totally absent in WFI as per IP **01**
- E Intrathecal route means administration to _____. **01**
- Q 5 Attempt any TWO of the following**
- A Write down merits and demerits of parenterals. **05**
- B
1. Differentiate the labelling requirement as mentioned below. **02**
1. Injection
2. For Injection
- Complete the given paragraph.
As per IP specification, Water for Injection is prepared by _____
2. method, Shall have an endotoxin levels of not more than _____ EU/ml **03**
and _____ cfu per 100ml.
- C Write a note on large volume parenterals and small volume parenterals. **05**
- Q 6 Attempt any TWO of the following**
- A Describe water and steam system for sterile production. **05**
- B Write a note on Form-Fill-Seal Technology. **05**
- C Enlist the content of sterile products manufacturing records **05**